

Quantitative Thinking #1

Why Is 45° So Important?

Date

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Project Goal

I wanted to understand how launch angle affects the distance a projectile can travel and why 45° is often considered the most important launch angle in projectile motion.

What I Worked On

- Researched the fundamentals of projectile motion.
- Learned how launch angle influences horizontal range.
- Studied the projectile range equation.
- Created an Excel spreadsheet to compare different launch angles.
- Calculated projectile ranges using a constant launch velocity.
- Organized the results into a data table.
- Created a graph to visualize the relationship between launch angle and range.

What I Learned

Before starting this analysis, I assumed that increasing the launch angle would always increase the distance traveled.

However, the results showed that range increases only up to a certain point and reaches its maximum at 45°.

I was also surprised to see that complementary angles such as 40° and 50° produced the same range. The same pattern appeared for 30° and 60°, and for 20° and 70°.

This analysis helped me better understand how mathematics can be used to predict real-world flight behavior.

Results

- Maximum range occurred at 45°.
- Range increased gradually between 10° and 45°.
- Range decreased after 45°.

- The graph showed a clear symmetrical pattern.
- The calculations matched the patterns shown in the graph.

Challenges

One of the most interesting parts of this analysis was understanding why different launch angles could produce identical ranges.

At first, I expected every angle to produce a unique result, but the mathematical relationship behind projectile motion explained the symmetry.

Key Lessons Learned

- Small changes in launch angle can significantly affect flight performance.
- Mathematics can be used to model engineering systems.
- Projectile motion follows predictable patterns.
- Data visualization makes engineering concepts easier to understand.
- Engineering often begins with asking simple questions.

Final Reflection

This analysis started with a simple question: Why is 45° so important?

Although the calculations were straightforward, the results showed how mathematics can reveal patterns that are not immediately obvious. Seeing the relationship between launch angle and range helped me better understand one of the basic concepts behind rocket and aircraft flight.

What I enjoyed most was seeing how a simple Excel model could help explain real-world motion through data and mathematics.

Small changes can completely change a mission.